

# SOP-INS-020 Heat Exchanger Shell Thickness Inspection

Revision B - Owner: Technical Services - Classification: Internal

## 1. Scope

Applies to shell-and-tube heat exchangers in hydrocarbon service. Condition monitoring locations (CMLs) are fixed points on the shell and nozzles where wall thickness is measured by ultrasonic testing (UT) at every inspection.

## 2. Corrosion rate

The short-term corrosion rate (CR) at a CML is the loss in thickness divided by the time between the two most recent readings:  $CR = (t_{\text{previous}} - t_{\text{actual}}) / \text{years between readings}$ , in mm/year. Use the highest CR among the CMLs of a component for remaining-life calculations.

## 3. Remaining life

Remaining life (RL) in years =  $(t_{\text{actual}} - t_{\text{min}}) / CR$ , where  $t_{\text{min}}$  is the minimum required thickness from the design calculation. If CR is zero or negative, record RL as greater than 20 years.

## 4. Next inspection date

The next internal inspection shall be scheduled at the lesser of half the remaining life or 10 years from the current inspection. Where the remaining life is less than 4 years the equipment shall be referred to the Integrity Review Board.

## 5. Reporting

The inspection report shall list every CML with its previous and actual thickness, the calculated corrosion rate, remaining life and the recommended next inspection date. Any CML below  $t_{\text{min}}$  requires an immediate fitness-for-service review.